

UAS PRAKTIKUM
WEB PROGRAMMING ADVANCE

“Disusun untuk memenuhi salah satu tugas uas praktikum mata kuliah web programing
advance”

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FAKULTAS ILMU TERAPAN DAN SAINS
INSTITUT PENDIDIKAN INDONESIA
GARUT 2025

Soal!!

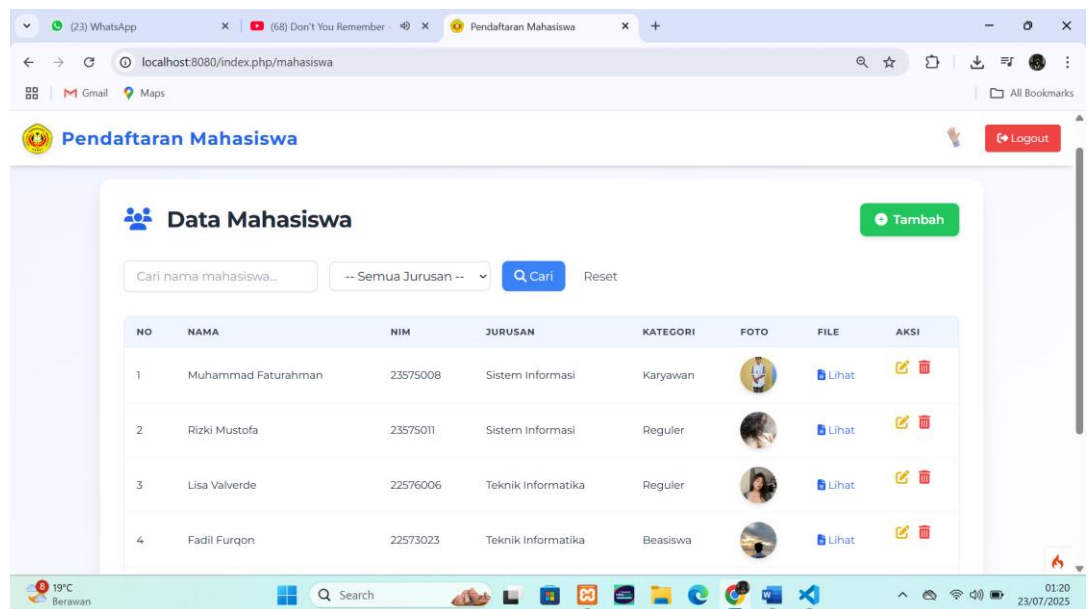
1. Buatlah CRUD sederhana untuk pendaftaran Mahasiswa!
2. Buatlah form handling untuk validasi data mahasiswa
3. Buatlah fitur upload file dan gunakan validasi (PDF, DOCX, JPG, PNG)
4. Buatlah sebuah aplikasi sederhana menggunakan Advanced Query

Jawaban!!

1. CRUD Pendaftaran Mahasiswa:

a. Hasil Tampilan

1) Index



2) Create

The screenshot shows a web browser window with the URL `localhost:8080/mahasiswa/create`. The page title is "Pendaftaran Mahasiswa". The main content is a form titled "Tambah Mahasiswa" with the subtitle "Isi data mahasiswa dengan benar". The form fields are as follows:

- Nama:** Arman Hardiansyah
- NIM:** 23575010
- Jurusan:** Teknik Informatika
- Kategori:** Mahasiswa (dropdown menu)
- Foto:** No file chosen (with a "Choose file" button)
- Upload File Baru (PDF, DOCX, 3PG, PNG):** No file chosen (with a "Choose file" button)

At the bottom of the form are two buttons: "Simpan Data" (Save Data) and "Kembali" (Back). The footer of the page reads "© 2025 Sistem Pendaftaran Mahasiswa | Institut Pendidikan Indonesia Garut".

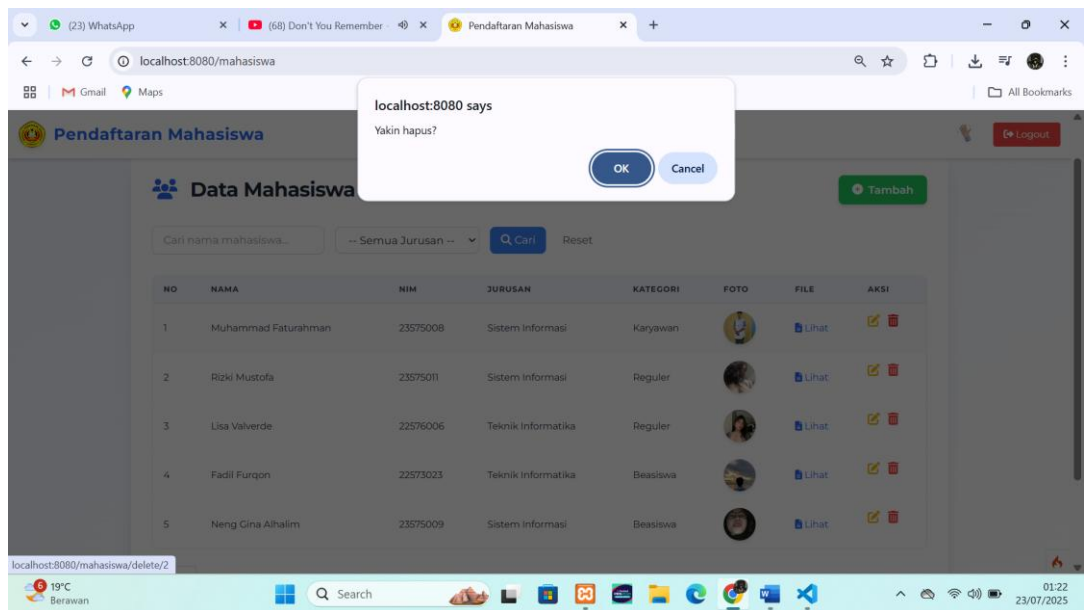
3) Edit

The screenshot shows a web browser window with the URL `localhost:8080/mahasiswa/edit/2`. The page title is "Pendaftaran Mahasiswa". The main content is a form titled "Edit Mahasiswa" with the subtitle "Perbarui data mahasiswa dengan benar". The form fields are as follows:

- Nama:** Muhammad Faturrahman
- NIM:** 23575008
- Jurusan:** Sistem Informasi
- Kategori:** Karyawan (dropdown menu)
- Foto Saat Ini:** A small thumbnail image of a person in a white shirt and blue pants.
- Choose file:** No file chosen (with a "Choose file" button)
- Upload File Baru (PDF, DOCX, 3PG, PNG):** No file chosen (with a "Choose file" button)

At the bottom of the form are two buttons: "Update Data" and "Kembali" (Back).

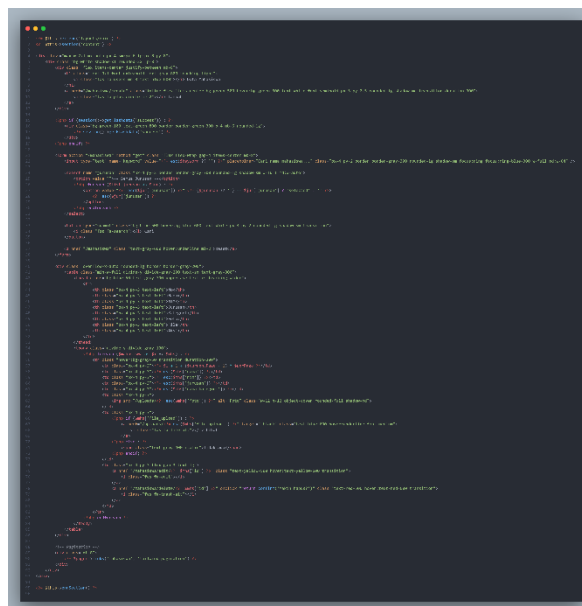
4) Delete



b. Listing Code

1) Index

a) Index.php



b) Controller Index Mahasiswa.php

2) Create

a) Create.php

b) Controller Create Mahasiswa.php

```
1 public function create()
2 {
3     $data['kategori'] = $this->kategoriModel->findAll();
4     return view('mahasiswa/create', $data);
5 }
6
7 public function store()
8 {
9     $validation = \Config\Services::validation();
10    $data = $this->request->getPost();
11
12    $rules = [
13        'nama' => 'required|min_length[3]',
14        'nim' => 'required|numeric|is_unique[mahasiswa.nim]',
15        'jurusan' => 'required',
16        'foto' => 'uploaded[foto][is_image[foto]]',
17        'file_upload' => 'uploaded[file_upload][ext_in[file_upload,pdf,docx,jpg,png]]'
18    ];
19
20    if (!$this->validate($rules)) {
21        return redirect()->back()->withInput()->with('errors', $validation->listErrors());
22    }
23
24    $foto = $this->request->getFile('foto');
25    $namaFoto = $foto->getRandomName();
26    $foto->move('uploads', $namaFoto);
27
28    $file = $this->request->getFile('file_upload');
29    $namaFile = $file->getRandomName();
30    $file->move('uploads', $namaFile);
31
32    $this->mahasiswaModel->save([
33        'nama' => $data['nama'],
34        'nim' => $data['nim'],
35        'jurusan' => $data['jurusan'],
36        'foto' => $namaFoto,
37        'file_upload' => $namaFile,
38        'kategori_id' => $data['kategori_id'] ?? null
39    ]);
40
41    return redirect()->to('/mahasiswa')->with('success', 'data berhasil ditambahkan');
42 }
```

3) Edit

a) Edit.php

```
1 public function edit($id)
2 {
3     $mahasiswa = $this->mahasiswaModel->find($id);
4     if (!$mahasiswa) {
5         return redirect()->to('/mahasiswa')->with('error', 'data tidak ditemukan');
6     }
7     $data['kategori'] = $this->kategoriModel->findAll();
8     return view('mahasiswa/edit', compact('mahasiswa', 'data'));
9 }
10
11 public function update($id)
12 {
13     $mahasiswa = $this->mahasiswaModel->find($id);
14     if (!$mahasiswa) {
15         return redirect()->to('/mahasiswa')->with('error', 'data tidak ditemukan');
16     }
17     $validation = \Config\Services::validation();
18     $data = $this->request->getPost();
19
20     $rules = [
21         'nama' => 'required|min_length[3]',
22         'nim' => 'required|numeric|is_unique[mahasiswa.nim]',
23         'jurusan' => 'required',
24         'foto' => 'uploaded[foto][is_image[foto]]',
25         'file_upload' => 'uploaded[file_upload][ext_in[file_upload,pdf,docx,jpg,png]]'
26     ];
27
28     if (!$this->validate($rules)) {
29         return redirect()->back()->withInput()->with('errors', $validation->listErrors());
30     }
31
32     $foto = $this->request->getFile('foto');
33     $namaFoto = $foto->getRandomName();
34     $foto->move('uploads', $namaFoto);
35
36     $file = $this->request->getFile('file_upload');
37     $namaFile = $file->getRandomName();
38     $file->move('uploads', $namaFile);
39
40     $this->mahasiswaModel->update($id, [
41         'nama' => $data['nama'],
42         'nim' => $data['nim'],
43         'jurusan' => $data['jurusan'],
44         'foto' => $namaFoto,
45         'file_upload' => $namaFile
46     ]);
47
48     return redirect()->to('/mahasiswa')->with('success', 'data berhasil diupdate');
49 }
50
51 public function delete($id)
52 {
53     $mahasiswa = $this->mahasiswaModel->find($id);
54     if (!$mahasiswa) {
55         return redirect()->to('/mahasiswa')->with('error', 'data tidak ditemukan');
56     }
57     $this->mahasiswaModel->delete($id);
58     return redirect()->to('/mahasiswa')->with('success', 'data berhasil dihapus');
59 }
60
61 public function index()
62 {
63     $data['kategori'] = $this->kategoriModel->findAll();
64     $mahasiswa = $this->mahasiswaModel->findAll();
65     return view('mahasiswa/index', compact('data', 'mahasiswa'));
66 }
67
68 public function show($id)
69 {
70     $mahasiswa = $this->mahasiswaModel->find($id);
71     if (!$mahasiswa) {
72         return redirect()->to('/mahasiswa')->with('error', 'data tidak ditemukan');
73     }
74     return view('mahasiswa/show', compact('mahasiswa'));
75 }
76
77 public function search()
78 {
79     $keyword = $this->request->get('keyword');
80     $mahasiswa = $this->mahasiswaModel->search($keyword);
81     return view('mahasiswa/search', compact('mahasiswa'));
82 }
83
84 public function filter()
85 {
86     $keyword = $this->request->get('keyword');
87     $jurusan = $this->request->get('jurusan');
88     $mahasiswa = $this->mahasiswaModel->search($keyword, $jurusan);
89     return view('mahasiswa/filter', compact('mahasiswa'));
90 }
91
92 public function pagination()
93 {
94     $keyword = $this->request->get('keyword');
95     $jurusan = $this->request->get('jurusan');
96     $mahasiswa = $this->mahasiswaModel->search($keyword, $jurusan);
97     return view('mahasiswa/pagination', compact('mahasiswa'));
98 }
99
100 public function export()
101 {
102     $keyword = $this->request->get('keyword');
103     $jurusan = $this->request->get('jurusan');
104     $mahasiswa = $this->mahasiswaModel->search($keyword, $jurusan);
105     return view('mahasiswa/export', compact('mahasiswa'));
106 }
107
108 public function import()
109 {
110     $file = $this->request->getFile('file_upload');
111     $namaFile = $file->getRandomName();
112     $file->move('uploads', $namaFile);
113     return view('mahasiswa/import', compact('namaFile'));
114 }
115
116 public function dashboard()
117 {
118     return view('mahasiswa/dashboard');
119 }
120
121 public function logout()
122 {
123     session_destroy();
124     return redirect()->to('/login');
125 }
```

b) Controller Edit Mahasiswa.php

```
1 public function edit($id)
2 {
3     $mahasiswa = $this->mahasiswaModel->find($id);
4     $kategori = $this->kategoriModel->findAll();
5
6     return view('mahasiswa/edit', [
7         'mahasiswa' => $mahasiswa,
8         'kategori' => $kategori
9     ]);
10 }
11
12 public function update($id)
13 {
14     $mahasiswa = $this->mahasiswaModel->find($id);
15     $validation = $this->validation->validate($id);
16
17     $rules = [
18         'nama' => 'required|min_length(3)',
19         'nim' => 'required|numeric',
20         'jurusan' => 'required',
21         'file_upload' => 'permit|image|max(1024000)'
22     ];
23
24     if (!$this->validation->validate($id)) {
25         return redirect()->back()->withInput()->with('errors', $validation->errors());
26     }
27
28     $data = [
29         'nama' => $this->request->getPost('nama'),
30         'nim' => $this->request->getPost('nim'),
31         'jurusan' => $this->request->getPost('jurusan'),
32         'kategori_id' => $this->request->getPost('kategori_id') // null
33     ];
34
35     $foto = $this->request->getPost('foto');
36     if ($foto && $foto->isValid() && $foto->hasMoved()) {
37         $namaFoto = $foto->getRandomName();
38         $foto->move('uploads', $namaFoto);
39         if (file_exists('uploads/' . $mahasiswa['foto'])) {
40             unlink('uploads/' . $mahasiswa['foto']);
41         }
42         $data['foto'] = $namaFoto;
43     }
44
45     $file = $this->request->getPost('file_upload');
46     if ($file && $file->isValid() && $file->hasMoved()) {
47         $namaFile = $file->getRandomName();
48         $file->move('uploads', $namaFile);
49         if (file_exists('uploads/' . $mahasiswa['file_upload'])) {
50             unlink('uploads/' . $mahasiswa['file_upload']);
51         }
52         $data['file_upload'] = $namaFile;
53     }
54
55     $this->mahasiswaModel->update($id, $data);
56     return redirect()->to('/mahasiswa')->with('success', 'Data berhasil diupdate');
57 }
```

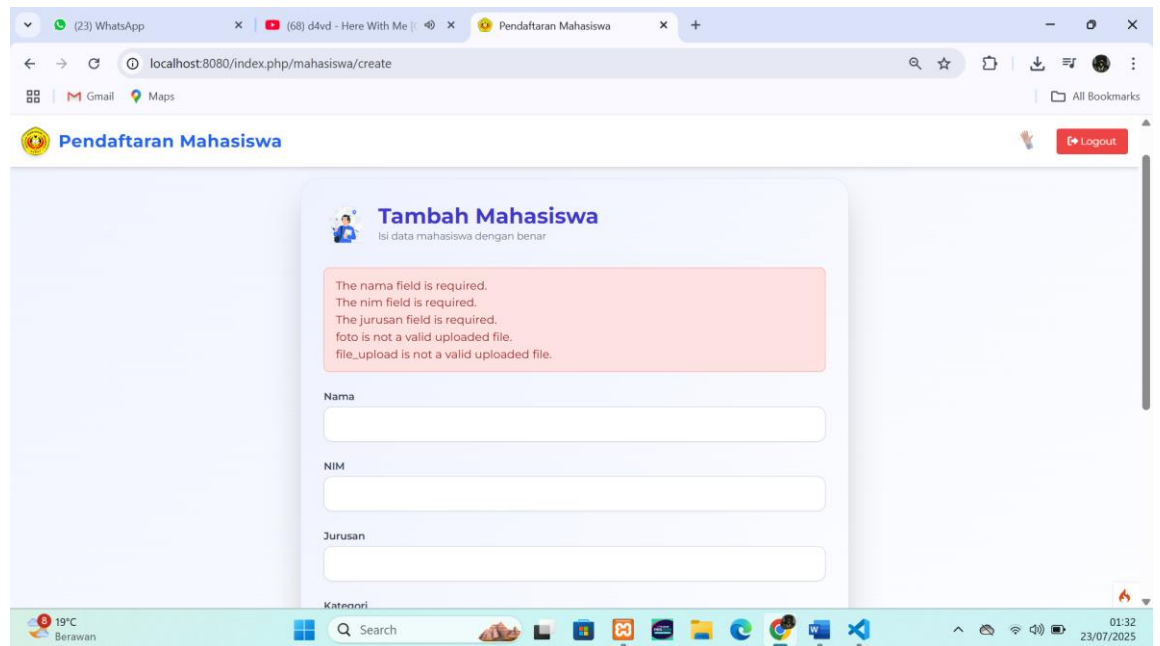
4) Delete

a) Controller Delete Mahasiswa.php

```
1 public function delete($id)
2 {
3     $mahasiswa = $this->mahasiswaModel->find($id);
4
5     if ($mahasiswa) {
6         if (file_exists('uploads/' . $mahasiswa['foto'])) {
7             unlink('uploads/' . $mahasiswa['foto']);
8         }
9         if (file_exists('uploads/' . $mahasiswa['file_upload'])) {
10             unlink('uploads/' . $mahasiswa['file_upload']);
11         }
12     }
13
14     $this->mahasiswaModel->delete($id);
15     return redirect()->to('/mahasiswa')->with('success', 'Data berhasil dihapus.');
```

2. Form Handling + Validasi Data

a. Hasil Tampilan



Note: Seperti tampilan diatas bahwa form create maupun edit sudah menggunakan form handling + validasi, dimana ketika kita tidak mengisi data ada pesan eror dimana nama, nim, jurusan, foto dibutuhkan. Dan nim harus berupa angka jangan huruf.

b. Listing Code

1) Form Handling + Validasi Form Create

```
1 public function create()
2 {
3     $data['kategori'] = $this->kategoriModel->findAll();
4     return view('mahasiswa/create', $data);
5 }
6
7 public function store()
8 {
9     $validation = \Config\Services::validation();
10    $data = $this->request->getPost();
11
12    $rules = [
13        'nama' => 'required|min_length[3]',
14        'nim' => 'required|numeric|is_unique[mahasiswa.nim]',
15        'jurusan' => 'required',
16        'foto' => 'uploaded[foto]|is_image[foto]',
17        'file_upload' => 'uploaded[file_upload]|ext_in[file_upload,pdf,docx,jpg,png]';
18    ];
19
20    if (!$this->validate($rules)) {
21        return redirect()->back()->withInput()->with('errors', $validation->listErrors());
22    }
23
24    $foto = $this->request->getFile('foto');
25    $namaFoto = $foto->getRandomName();
26    $foto->move('uploads', $namaFoto);
27
28    $file = $this->request->getFile('file_upload');
29    $namaFile = $file->getRandomName();
30    $file->move('uploads', $namaFile);
31
32    $this->mahasiswaModel->save([
33        'nama' => $data['nama'],
34        'nim' => $data['nim'],
35        'jurusan' => $data['jurusan'],
36        'foto' => $namaFoto,
37        'file_upload' => $namaFile,
38        'kategori_id' => $data['kategori_id'] ?? null
39    ]);
40
41    return redirect()->to('/mahasiswa')->with('success', 'Data berhasil ditambahkan');
42 }
```

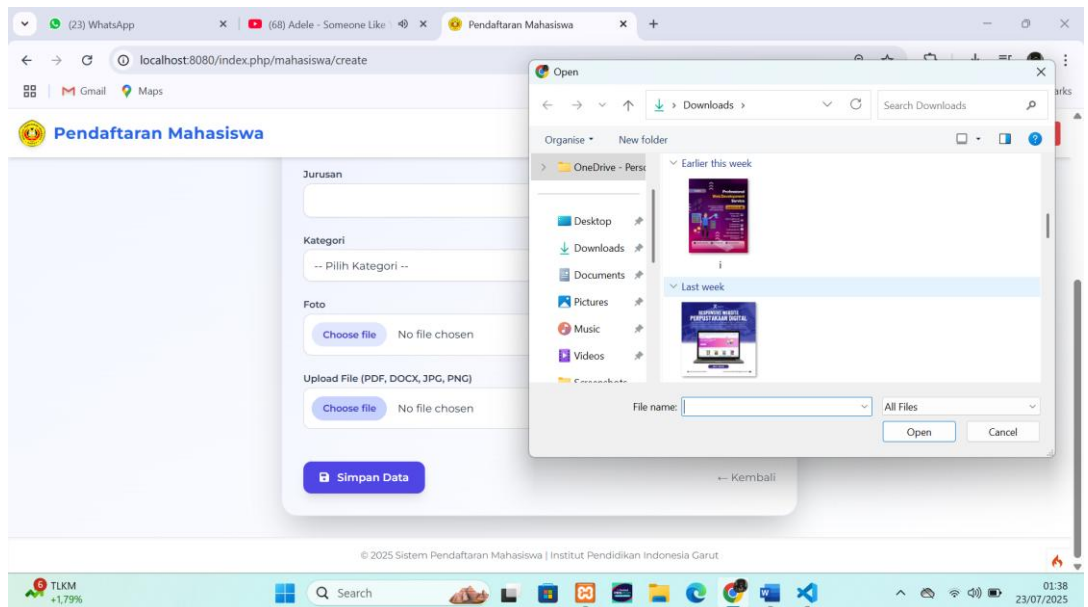

2) Form Handling + Validasi Form Edit

```
1 public function edit($id)
2 {
3     $mahasiswa = $this->mahasiswaModel->find($id);
4     $kategori = $this->kategoriModel->findAll();
5
6     return view('mahasiswa/edit', [
7         'mahasiswa' => $mahasiswa,
8         'kategori' => $kategori
9     ]);
10 }
11
12 public function update($id)
13 {
14     $mahasiswa = $this->mahasiswaModel->find($id);
15     $validation = $this->validation->validation();
16
17     $rules = [
18         'nama' => 'required|min_length(3)',
19         'nim' => 'required|numeric',
20         'jurusan' => 'required',
21         'file_upload' => 'permit|upload|max_size[1024000]|mime_image'
22     ];
23
24     if (!$this->validation->validate($validation)) {
25         return redirect()->back()->withInput()->with('errors', $validation->errors());
26     }
27
28     $data = [
29         'nama' => $this->request->getPost('nama'),
30         'nim' => $this->request->getPost('nim'),
31         'jurusan' => $this->request->getPost('jurusan'),
32         'kategori_id' => $this->request->getPost('kategori_id') // null
33     ];
34
35     $foto = $this->request->getFile('foto');
36     if ($foto && $foto->isValid() && $foto->hasMoved()) {
37         $namaFoto = $foto->getRandomName();
38         $foto->move('uploads', $namaFoto);
39         if ($file_exists('uploads/' . $mahasiswa['foto'])) {
40             unlink('uploads/' . $mahasiswa['foto']);
41         }
42         $data['foto'] = $namaFoto;
43     }
44
45     $file = $this->request->getFile('file_upload');
46     if ($file && $file->isValid() && $file->hasMoved()) {
47         $namaFile = $file->getRandomName();
48         $file->move('uploads', $namaFile);
49         if ($file_exists('uploads/' . $mahasiswa['file_upload'])) {
50             unlink('uploads/' . $mahasiswa['file_upload']);
51         }
52         $data['file_upload'] = $namaFile;
53     }
54
55     $this->mahasiswaModel->update($id, $data);
56     return redirect()->to('/mahasiswa')->with('success', 'Data berhasil diupdate');
57 }
```

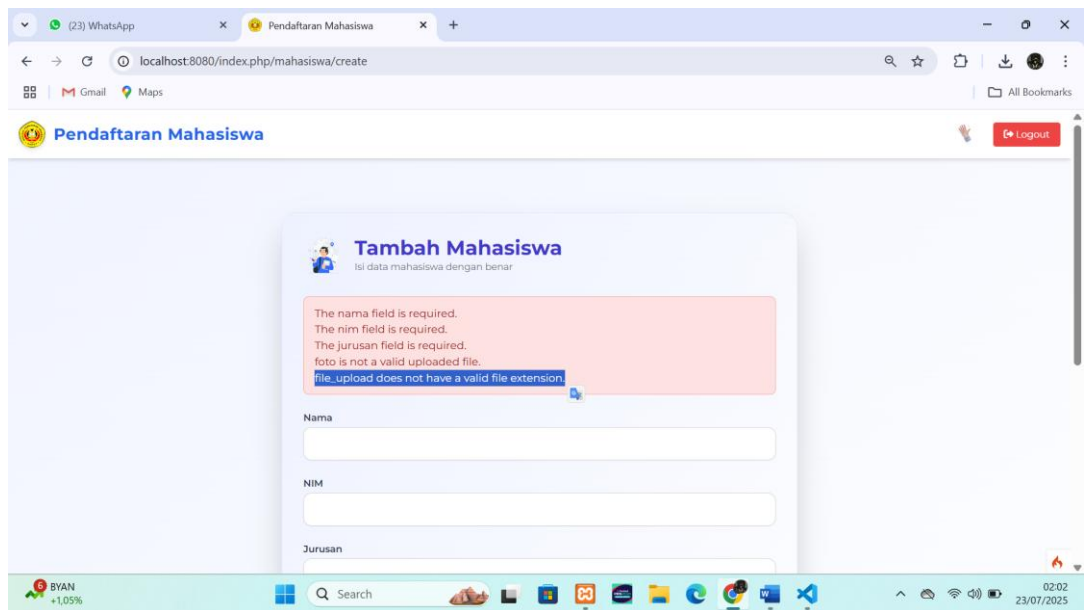
3. Fitur Upload File + Validasi Ektensi File

a. Hasil Tampilan

1) Upload File



2) Validasi File Ektensi



Note: Bisa dilihat seperti digambar bahwa ketika upload file selain ekstensi yang sudah di validasi, maka akan eror dan muncul pesan gagal.

b. Listing Code

```
1  $foto = $this->request->getFile('foto');
2  $namaFoto = $foto->getRandomName();
3  $foto->move('uploads', $namaFoto);
4
5  $file = $this->request->getFile('file_upload');
6  $namaFile = $file->getRandomName();
7  $file->move('uploads', $namaFile);
8
9  $this->mahasiswaModel->save([
10     'nama' => $data['nama'],
11     'nim' => $data['nim'],
12     'jurusan' => $data['jurusan'],
13     'foto' => $namaFoto,
14     'file_upload' => $namaFile,
15     'kategori_id' => $data['kategori_id'] ?? null
16  ]);
```

4. Advance Query

a. Hasil Tampilan

1) Join antar table (kategori)

The screenshot shows a web browser window with the URL `localhost:8080/mahasiswa/create`. The page title is "Pendaftaran Mahasiswa". The form contains the following fields:

- Jurusan**: A text input field.
- Kategori**: A dropdown menu with the following options: `-- Pilih Kategori --`, `-- Pilih Kategori --`, **Beasiswa** (selected), `Reguler`, and `Karyawan`.
- Upload File (PDF, DOCX, JPG, PNG)**: A button labeled "Choose file" and the text "No file chosen".
- Buttons**: "Simpan Data" and "Kembali".

The footer of the page reads: "© 2025 Sistem Pendaftaran Mahasiswa | Institut Pendidikan Indonesia Garut".

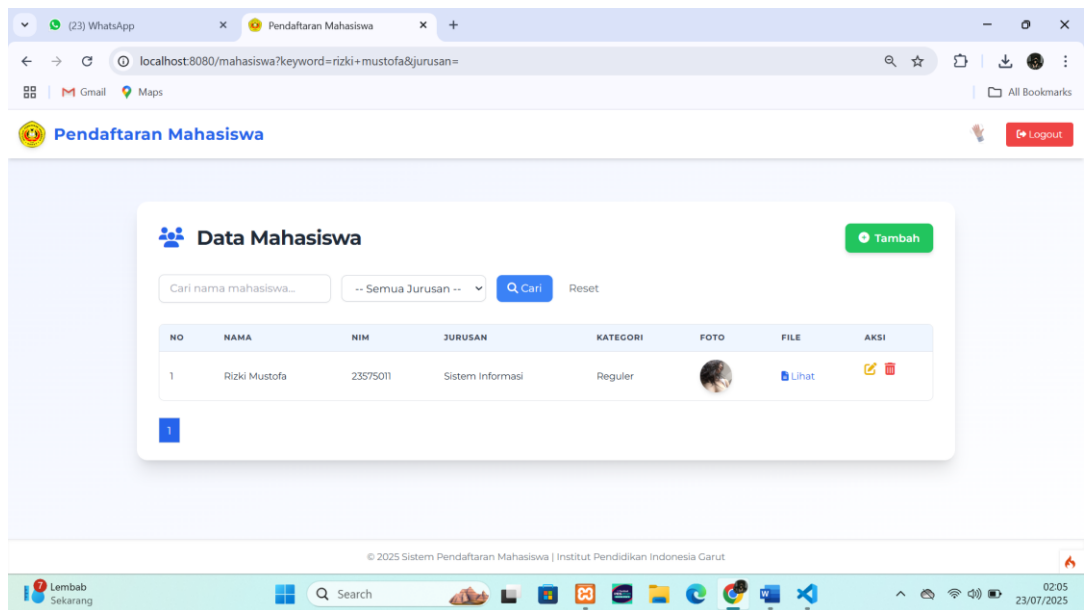
2) Filter Jurusan

The screenshot shows a web browser window with the URL `localhost:8080/mahasiswa?keyword=&jurusan=Sistem+Informasi`. The page title is "Pendaftaran Mahasiswa". The table displays the following data:

NO	NAMA	NIM	JURUSAN	KATEGORI	FOTO	FILE	AKSI
1	Muhammad Faturahman	23575008	Sistem Informasi	Karyawan		Lihat	Edit Hapus
2	Rizki Mustofa	23575011	Sistem Informasi	Reguler		Lihat	Edit Hapus
3	Neng Cina Alhalim	23575009	Sistem Informasi	Beasiswa		Lihat	Edit Hapus
4	Peri Latuyatim	22576004	Sistem Informasi	Reguler		Lihat	Edit Hapus

The footer of the page reads: "© 2025 Sistem Pendaftaran Mahasiswa | Institut Pendidikan Indonesia Garut".

3) Filter Nama



b. Listing Code

1) Join Antar Table



2) Filter Jurusan



3) Filter Nama



```
1 public function searchByName($keyword)
2     {
3         return $this->like('nama', $keyword)->findAll();
4     }
```